Singampalli Sri Varshith

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Education

Year	Degree/Certificate	Institute	CPI/%
2025 - Present	M.Tech(Computer Science and Engineering)	Indian Institute of Technology, Kanpur	8.55/10
2021 - 2025	B.Tech(Computer Science and Engineering)	Gayatri Vidya Parishad College of Engineering	8.96/10
2019 - 2021	Intermediate(XII)	Sri Viswa Junior College	95.2%
2018 - 2019	Matriculation(X)	Sri Chaitanya Techno School	89.4%

Research Experience

- **Reliable and Interpretable Medical AI Tool Orchestration for Cardiac MRI** (Ongoing) *(Jul'26 – Present)*
  Guide: Prof. Arnab Bhattacharya, Department of Computer Science and Engineering, IIT Kanpur
  - Curating ground-truth tool-call trajectories for cardiac MRI questions by annotating required, optional, and irrelevant tools along with the expected tool-calling sequence and inputs, to enable systematic evaluation of agent orchestration.
  - Building a tool-calling evaluation harness measuring tool-selection precision/recall/F1, sequence correctness, and unnecessary/missed tool rates beyond final-answer accuracy.
  - Exploring multi-level uncertainty estimation across tool selection, individual tool outputs, workflow-level call sequences, and final clinical answers to quantify agent confidence and improve interpretability and reliability of medical tool orchestration.

Key Projects

- **Code-Aware RAG for Python**(Self Project) *(Aug'26)*
  - Developed a **code-aware RAG system** for Python repositories that retrieves relevant code and generates answers with **file:line citations** for source traceability.
  - Used **tree-sitter** for structure-aware code chunking and combined **semantic and keyword retrieval** to handle conceptual queries and exact code identifiers.
  - Evaluated retrieval using **Recall@K, MRR, and nDCG**, analyzing reranking trade-offs and removing cross-encoder reranking due to increased latency and reduced retrieval performance.
  - Implemented **abstention for unsupported queries**, achieving **30/30** correct answers on supported questions and **15/15** correct refusals on unanswerable questions.
- **Campus Resource Allocation System** (Self Project) *(Aug'26)*
  - Built an allocation backend for GPU clusters, rooms, and course seats, enforcing capacity and per-role quota as **separate serialization points** — resource-row and user-row locks acquired in a fixed order (SELECT FOR UPDATE) so deadlock cannot form — proven with an asyncio harness that drops over-allocation from X to zero under 500 concurrent requests for 50 seats.
  - Guaranteed **exactly-once allocation** across client retries by committing a unique idempotency key in the same transaction as the booking itself, alongside JWT auth with role-based route gating, Postgres GiST exclusion constraints for overlap-free room intervals, and a FIFO waitlist with quota-aware promotion under concurrent drops.
- **RescueSeg: Hybrid-Attention Transformer for Segmentation** (Course Project: CS776) *(Apr'26)*
  - Co-developed a hybrid Transformer encoder combining **Overlapping Cross Attention** and **Global Sparse Attention** with a learnable fusion gate, FPN, and Attention U-Net decoder.
  - Added a **boundary-refinement module** to sharpen object boundaries and improve segmentation of fine-grained damage regions.
  - Achieved **77.35% mIoU** with **44 ms** inference on RescueNet across 11 classes, outperforming a Segmenter baseline under identical settings.
  - Conducted a **3-variant ablation** on boundary refinement and auxiliary supervision, identifying the components responsible for improved boundary quality and stable training.
  - Co-developed a **hybrid Transformer-based segmentation model** combining local and global attention with FPN and Attention U-Net for disaster-damage analysis.
  - Designed a **boundary-refinement module** and auxiliary supervision strategy to improve segmentation of fine-grained damage regions and object boundaries.
  - Achieved **77.35% mIoU** on RescueNet across 11 classes with **44 ms inference**, outperforming a Segmenter baseline.
- **Dual-Discriminator GAN with Self-Attention** (Course Project: CS787) *(Sep'25–Nov'25)*
  - Designed and implemented a **Dual Discriminator GAN (D2GAN)** with self-attention for CIFAR-10 image synthesis using custom generator and discriminator architectures.
  - Incorporated transposed convolutions, spectral normalization, batch normalization, and a dual-discriminator objective to improve image realism, diversity, convergence, and **reduce mode collapse.**
  - Benchmarked generative quality using **Fréchet Inception Distance (FID)** and **Inception Score (IS)**, with checkpointing to analyze training performance across models.
  - Designed and implemented a **Dual Discriminator GAN** with self-attention for CIFAR-10 image generation using custom generator and discriminator architectures.
  - Applied **spectral normalization, batch normalization, and transposed convolutions** to improve training stability and generated image quality.
  - Evaluated generative performance using **FID and Inception Score (IS)**, analyzing model checkpoints to compare training progress and image synthesis quality.

- **Global Health Dashboard: Visual Analytics of Healthcare Inequality** (Course Project: CS661) (*Jun'26*)
  - Developed an interactive visual analytics dashboard analyzing **72,420 country–year–sex estimates** from WHO data to study global inequalities in diabetes and hypertension treatment and effective control.
  - Implemented **six linked visualizations** including choropleth maps, sex-gap analysis, temporal trends, treatment-to-control cascades, and region/income comparisons with cross-filtering and shared state.
  - Built a **single-page application** with centralized state management, nearest-year fallback, dynamic KPI computation, and reusable Plotly visualization modules for scalable exploratory analysis.
  - Developed an interactive **healthcare analytics dashboard** using WHO data to visualize global inequalities in diabetes and hypertension treatment and control.
  - Implemented **linked visualizations** including choropleth maps, temporal trends, sex-gap analysis, and treatment-to-control comparisons with interactive filtering.
  - Built a **single-page visual analytics application** with centralized state management, dynamic KPIs, and reusable Plotly-based visualization components.
- **Email Spam Detection System** (Self Project) (*Mar'26 - Apr'26*)
  - Developed a **transformer-based spam classification system** by fine-tuning **BERT** on a dataset of 5,500+ labeled messages, leveraging contextual language representations for accurate spam detection.
  - Built an end-to-end inference pipeline using **Hugging Face Transformers**, implementing tokenization, model fine-tuning, and probabilistic classification.
  - Developed a **BERT-based email spam classifier** by fine-tuning a pretrained Transformer model on 5,500+ labeled messages.
  - Built an end-to-end **text classification pipeline** using Hugging Face Transformers for tokenization, model training, and inference.
- **Huffman File Compressor** (Self Project) (*Apr'24*)
  - Implemented **lossless Huffman compression** using frequency analysis, binary trees, and prefix-free codes.
  - Built end-to-end **bitstream encoding/decoding** with Huffman tree serialization for lossless reconstruction.
  - Measured compression performance using **compression ratio** across different input files.
  - Implemented a **lossless Huffman compression** algorithm using frequency analysis, binary trees, and prefix-free encoding.
  - Built end-to-end **encoding and decoding** with Huffman tree serialization for accurate file reconstruction.
  - Evaluated compression efficiency using **compression ratios** across different input files.
- **Distributed 3D Stencil Solver with MPI Halo Exchange** (Course Assignment: CS633) (Apr'26)
  - Developed a parallel **3D stencil solver** in C using MPI, partitioning the grid across processes with iterative neighbour-boundary exchanges.
  - Organized processes on a **3D Cartesian topology** with six-neighbour ghost-cell halo exchange to maintain consistent boundary values.
  - Optimized communication using **MPI derived datatypes** for direct transfer of non-contiguous halo faces and non-blocking communication to overlap computation.
  - Achieved **1.17–1.20× speedup** on PARAM Rudra across 32–96 processes for a 240^3 grid, with communication overhead limiting gains at smaller problem sizes.
  - Developed a parallel **3D stencil solver** in C using MPI with domain decomposition and iterative neighbour-boundary exchanges.
  - Implemented a **3D Cartesian process topology** with six-neighbour halo exchange for efficient distributed boundary updates.
  - Optimized communication using **MPI derived datatypes and non-blocking communication**, achieving up to **1.20× speedup** across 32–96 processes.
- **Arbiter PUF Modeling Attack via Linear SVM** (Course Assignment: CS771) (*Aug'25 – Dec'25*)
  - Developed a **linear SVM attack** on Arbiter PUF challenge-response pairs using a delay-based feature transformation to linearize the underlying response function.
  - Implemented the **Stochastic Dual Coordinate Ascent (SDCA)** solver from scratch with primal-dual objective tracking and convergence monitoring.
  - Achieved **95.9% test accuracy** using only 1,000 of 100K challenge-response pairs; identified and fixed **4 solver-update bugs**.
  - Developed a **linear SVM-based modeling attack** on Arbiter PUFs using delay-based feature transformation of challenge-response pairs.
  - Implemented the **SDCA optimization algorithm from scratch** with primal-dual objective tracking and convergence monitoring.
  - Achieved **95.9% test accuracy** using only 1,000 challenge-response pairs and debugged the solver implementation for reliable convergence.
- **EM-Based Generative Classification with Gaussian Mixtures** (Course Assignment: CS771) (*Aug'25 – Dec'25*)
  - Implemented **class-conditional GMMs** and the EM algorithm from scratch to model MNIST digit distributions and perform generative classification.
  - Performed **MAP classification** with numerically stable log-sum-exp likelihood computation and extended the model to reconstruct images with missing pixels.
  - Evaluated mixture sizes from **K = 1–20**, finding accuracy saturation around K = 10–15 and using class-specific K values to match the K = 20 baseline with fewer parameters.
  - Implemented **Gaussian Mixture Models (GMMs)** and the Expectation-Maximization (EM) algorithm from scratch for generative classification on the MNIST dataset.
  - Developed a **MAP-based classification framework** with numerically stable likelihood computation and image reconstruction for missing-pixel scenarios.
  - Analyzed the impact of **mixture size selection** on classification performance, balancing model accuracy and parameter efficiency.
- **Email Spam Classification** (Self Project) (*Dec'25 - Jan'26*)
  - Built an **NLP-based spam classifier** using TF-IDF features and Logistic Regression to distinguish spam from legitimate emails.
  - Implemented **text preprocessing, feature extraction, model training, and evaluation** using Scikit-learn and Pandas.
  - Developed a **web interface** for real-time email classification, reusing the trained TF-IDF vectorizer and model for inference.
  - Built an **NLP-based spam classifier** using TF-IDF features and Logistic Regression to classify emails as spam or legitimate.
  - Implemented an end-to-end **text classification pipeline** covering preprocessing, feature extraction, model training, and evaluation.

- Developed a **web interface for real-time inference**, integrating the trained TF-IDF vectorizer and classification model.

Technical Skills and Courses

- **Languages:** C, C++, Python, SQL
- **Libraries:** Scikit-learn, Pytorch, NumPy, Pandas, Matplotlib
- **Frameworks and Tools:** fastapi, Git, Github, Docker, Postman, Langchain, plotly, Streamlit, LangGraph
- **Courses:** Introduction to ML, Generative AI, Deep learning for Computer Vision, Design and Analysis of Algorithms, Big Data Visual Analytics

Scholastic Achievements

- Selected for **ICPC (International Collegiate Programming Contest, Amritapuri Region)** Regionals, 2024.
- Selected for **ICPC (International Collegiate Programming Contest, Amritapuri Region)** Regionals, 2023.
- Selected for **ICPC (International Collegiate Programming Contest, Kanpur Region)** Regionals, 2023.
- Achieved **2nd position** in Cook the Code Coding Contest, held in Gayatri Vidya Parishad College of Engineering.
- Achieved **2nd position** in IBC Alt Hack, Blockchain based Ideathon.

Coding Profiles

- **Codeforces** – 1410 (Specialist, Max Rating)
- **Codechef** – 1712 (Max Rating)
- **Leetcode** – 1785 (Max Rating)

Positions of Responsibility

- **Student Guide, IIT Kanpur** (*2026-present*)
  Guided students on academic, administrative, and campus-related matters, facilitating access to appropriate resources and support.Assisted in resolving student queries by coordinating with faculty, academic offices, and relevant campus services. Supported effective student communication, mentoring, and orientation within the IIT Kanpur community.
- **Event Management Lead, Google Developer Student Club** (*2023-24*)
  Been a part of the core team of Google Developer Student Club, Gayatri Vidya Parishad College of Engineering, Visakhapatnam. Served as an Event Management Lead and organized coding contests, speaker sessions and lead workshops.